Arnab Kumar Chand

Machine Learning Engineer

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SUMMARY

- Around 5 years of experience as a Machine Learning Engineer specializing in natural language processing (NLP), Artificial Intelligence, Deep Learning, Machine Learning, and Data Mining, GenAI (LLM), and computer vision.
- Expertise in building various machine learning models using algorithms such as Linear Regression, Logistic Regression, Naive Bayes, Support Vector Machines (SVM), Decision trees, KNN, K-means Clustering, and Ensemble methods (Bagging, Gradient Boosting).
- Proficient in implementing and fine-tuning deep learning algorithms, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), LSTM, Transformers, BERT, GPT, and Llama.
- Excellent in Python to manipulate data for data loading and extraction and worked with Python libraries like NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow, Ggplot2, OpenCV, PyTorchand NLTK.
- Expert in managing AWS cloud resources like EC2, S3, Elastic Load Balancer, RDS, Glacier, SQS, SNS, Cloud Formation, Route53 and Identity and Access Management.

EDUCATION

Master of Science in Machine Learning | Stevens Institute of Technology, Hoboken, NJ

TECHNICAL SKILLS

- **Language:** Python, C++, SQL, R, SAS
- **Machine Learning:** Linear, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, XGBoost, K-Means
- **Deep Learning Framework:** Tensor-flow, Keras, PyTorch, ANN, CNN, RNN, LSTM, Attention and Transformers (BERT, GPT)
- **GenAI:** LangChain, LLaMA, Mistral, LlamaIndex, GANs, BERT, GPT-2 and 3
- **Cloud platforms:** GCP (Vertex AI, BigQuery, AutoML), AWS (SageMaker, S3, Glue), Azure
- **Libraries:** PySpark, OpenCV, Pillow, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, NLTK, Spacy, Seaborn
- **Visualizations and Tools:** Tableau, Power BI, Git, Bitbucket, Docker, Kubernetes, OpenVino
- **Database:** SQL Server, MySQL, PostgreSQL, MongoDB, Firebase
- **Operating System:** Windows, Linux, IOS

PROFESSIONAL EXPERIENCE

Capital One Financial, NJ, USA — ***Oct 2024 – Present***

Machine Learning Engineer

- Conducted hyperparameter tuning using GridSearchCV to optimize model performance, resulting in a 5% improvement in F1-score.
- Designed and implemented a high-performance backend using FastAPI and WebSockets to integrate the machine learning inference pipeline. Leveraged multiprocessing to achieve a 2x speed increase, and implemented robust monitoring features including real-time performance tracking, system resource monitoring, and detailed logging.
- Collaborated on developing a customer service chatbot using Large Language Models (LLMs), integrating with Confluence to retrieve and parse knowledge base articles, enhancing customer satisfaction and service efficiency by 25%.
- Improved model performance by an average of 10% by fine-tuning BERT for text classification and incorporating data generated by Generative Adversarial Networks (GANs).
- Built and deployed a fraud detection system using LSTM networks, identifying 200 fraudulent transactions with a 98% accuracy rate.

GoQuant, New Jersey (Remote), USA — ***Jul 2024 – Oct 2024***

Machine Learning Engineer

- Designed and implemented a chatbot for the platform, fine-tuning FinLLaMA on custom crypto data to handle real-time user queries and enable high-frequency trading, allowing users to place trade orders via prompts.
- Fine-tuned a BERT model for intent classification and Named Entity Recognition (NER), achieving 93.6% accuracy in extracting key entities from user prompts.
- Developed a Retrieval-Augmented Generation (RAG) pipeline using vector similarity with pgvector and PostgreSQL, optimizing trading data retrieval and improving response speed by 20%.

BigCity Promotions, Bangalore, India — ***Aug 2020 – Aug 2022***

AI & Deep Learning Developer

- Developed an automated invoice analysis using Optical Character Recognition (OCR) and RNN based Named Entity Recognition (NER) algorithms for detecting key-value pairs, table structure, and line items with an accuracy of 92%
- Boosted ML module's efficiency by 38% through algorithm optimization, model quantization, and pruning.
- Analyzed raw invoice data to identify customer purchase patterns during promotional events. Conducted extensive data cleaning and feature engineering to prepare the dataset for analysis. Employed pandas, matplotlib, and seaborn to visualize key trends and insights. Discovered a significant 20% increase in demand for promoted products, among other valuable findings.
- Trained XGBoost model to identify products and promotions correlated with higher customer purchases during promotional seasons. Model insights helped create an optimized promotion strategy for peak season, leading to 10% increase in demand for promoted products.
- Utilized Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, and NLTK in Python for developing various machine-learning algorithms.

Xarpie Labs, Bangalore, India ***Dec 2019 – Aug 2020***

Associate Engineer

- Conducted a comprehensive EDA on social media posts, extracting text, metadata, and NLP features. Developed a Long Short-Term Memory (LSTM) model for multi-class sentiment prediction, achieving an 89% validation F1-score.
- Deployed a Sanic API to operationalize the sentiment prediction model, seamlessly integrating it with a social media analytics platform. This enabled real-time insights into brand sentiment and key reputation drivers, providing valuable information for decision-making.
- Leveraged Tekton pipelines with GitOps workflow, enabling faster rollbacks and deployments for ML models.
- Streamlined the ML model deployment process by integrating AWS Code Pipeline, achieving a 50% reduction in deployment time and supporting continuous integration/continuous delivery (CI/CD) for machine learning projects.
- Engineered scalable ETL pipelines using PySpark to process 1M+ records from SQL Server and PostgreSQL sources, implementing distributed data processing pipelines for cleaning, transformation, and validation.
- Optimized performance through DataFrame operations and SQL functions, reducing processing time by 40% while ensuring data quality and consistency for analysis.

Wesense.ai, Bangalore, India **Oct *2018 – Nov 2019***

AI Scientist

- Utilized time-series analysis to identify and predict peak hours, enabling strategic staffing adjustments that resulted in a 30% reduction in operational costs during off-peak periods.
- Built deep learning models with PyTorch for diverse tasks like image recognition (object detection), NLP tasks (sentiment analysis, topic modeling), and time series forecasting, achieving validation accuracy of 85%.
- Implemented a comprehensive suite of data science and machine learning libraries such as Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, PyTorch, and NLTK to refine models and workflows, boosting development speed by 25%.
- Enhanced monitoring and operational excellence of ML systems by leveraging AWS CloudWatch, resulting in a 30% decrease in issue resolution times and a 56% uptime for critical ML workflows.

ACADEMICS

Research project – Accessible Image Search

- Designed and developed an iOS app for visually impaired users, leveraging a Flask backend for real-time object detection and scene analysis.
- Integrated the LLaVA vision-language model to generate precise text descriptions of objects and scenes, improving user accessibility by 40%.
- Delivered multi-modal results (audio and text) through the app, significantly enhancing user interaction with visual content.